

Editorial

Assessment of educational innovation¹

Molecular biologists and biochemists devote their professional lives to assessing the outcomes of their research and reporting that research in a manner that clearly documents those outcomes. It is time we adopted a similar attitude toward our efforts in undergraduate education.

We all have a reasonable idea of the type of information we feel necessary to assess the success of our own or our colleagues' research efforts. Molecular biologists and biochemists before beginning a research project: establish clear objectives; design, adopt, or adapt protocols to continually assess progress towards realizing those objectives; establish the validity of the protocols for the purpose used; and envision the nature of the data they hope to gather. The subsequent outcomes of these projects are easily followed through reports in the peer reviewed press. Acceptance of the advances reflects the care and clarity with which goals are defined and data unambiguously establishing the results of the research is presented.

We are less sure of how to proceed when faced with the need to establish the success of our own or our colleagues' "experiments" in undergraduate education. Often when scientists report education projects, either in peer reviewed journals or to the agency supporting their efforts, they discuss the needs they tried to address and the new approaches they introduced, but, because of the complexity of the goals and the difficulty of determining that they have been reached, documentation of outcomes is ignored or given only negligible attention. However, assessment and dissemination are crucial components of all projects. If we are to continue to generate a cadre of individuals interested and well trained in biochemistry and molecular biology and convince our colleagues to incorporate recent results of research into teaching and learning into their teaching styles, educational advances related to these disciplines need to be well documented and then rapidly disseminated so that others can benefit from the experiences of the pioneers in biochemistry/molecular biology education change.

¹The views expressed are those of the author and do not necessarily reflect those of the Division of Undergraduate Education, the Directorate for Education and Human Resources, or the National Science Foundation

The National Science Foundation, concerned that scientists have effective mechanisms for assessing the success of their efforts to improve undergraduate education, added two new assessment programs, *Assessment of Student Achievement in Undergraduate Education* (NSF 01-82) and *Research on Learning and Education* (NSF 00-17), to their broad array of undergraduate education funding programs. Both focus on appropriate design, adaptation and use of assessment tools to help science and mathematics faculty determine the results of their efforts to change undergraduate education. These programs are of particular importance now. Funding agencies are increasing the funds they invest to support curricular change in the sciences and mathematics. They are also requiring that proposals include descriptions of how the proposer will monitor progress in instituting the changes proposed and will document the outcome of these changes.

To scientists engaged in educational projects evaluation might seem a daunting task, but there is no need to become an educational assessment expert in order to design a good set of appropriate assessment tools. During the initial stages of planning a project, scientists should consult evaluators on their campus either in the psychology, education, anthropology or sociology department (changing classroom practices is a change in a culture). The web sites listed below also contain helpful information.

As successful projects begin to emerge everyone in the science community should be able to benefit from them, whether they be research or education related. Find a better way to clone or to carry out microbial synthesis of human proteins processed rapidly and with complete fidelity and the results are announced in a top banner headline in *Science* and *Nature* and even the *New York Times*. Just think how quickly the mysteries of such processes as apoptosis and gene duplication were disseminated and methods of studying them adopted. Let us hope a modicum of such interest can be generated for successful efforts to change the educational landscape.

References

- [1] <http://www.wcer.wisc.edu/nise/CL1/flag/> Field-Tested Learning Assessment Guide (FLAG) This web site has been designed by and for college and university instructors in the sciences, mathematics, engineering and technology (SMET) disciplines and contains some very useful surveys and guidelines created for chemistry that could be adapted to biochemistry.
- [2] <http://www.nsf.gov/pubs/20001/nsf0158/nsf0158.htm> NSF 00-117: Supplemental Information for Principal Investigators and Applicants to NSF's Course, Curriculum, and Laboratory

Improvement Program. Designed as an aid to funded projects, many of the suggestions have universal application.

- [3] <http://www.wkkf.org/Documents/WKKF/EvaluationHandbook/EvalHandbook.pdf> W.K. Kellog Foundation, Evaluation Handbook. Designed to serve administrators of systemic projects, this handbook also contains examples useful for individual faculty working on single course changes.

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